

# Future of AI, NN and DL in Pakistan

قومی اے آئی پالیسی 2025 کے مطابق مصنوعی ذہانت پاکستان کی منصوبہ بندی کا حصہ ہے: جدت، مہارت، اخلاقی استعمال اور ڈیجیٹل ترقی۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 07 of 17

## Future of AI, NN and DL in Pakistan

### LEARN IT

Machine learning, neural networks and deep learning will play a key role in Pakistan's future. The **National AI Policy 2025** highlights how artificial intelligence is now part of national planning, focusing on **innovation, skill-building, ethical use and digital growth**.

| Sector               | Potential impact in Pakistan                                                                                                |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Education            | Smart learning systems, online tutoring, automatic assessments, and finding where students need specific help.              |
| Healthcare           | Disease detection, medical image analysis, and managing hospital data. Crucial for rural areas lacking specialist doctors.  |
| Agriculture          | Monitoring crops, predicting weather, managing irrigation, and estimating yields — highly important for Pakistan's economy. |
| Business and banking | Fraud detection, customer support (chatbots), market forecasting, and intelligent risk analysis.                            |
| Local languages      | Speech recognition, translation and text-to-speech tools for <b>Urdu, Sindhi</b> and other regional languages.              |

To benefit from this future, Pakistan needs skilled individuals. You should focus on **programming (such as Python)**, **mathematics**, **data analysis**, **problem-solving** and the **ethical use of technology**.

### WORKED EXAMPLE

**Worked example:** pick a sector and write a three-line "impact paragraph" for 5 marks.

**Agriculture:** A cotton grower in Sanghar photographs leaves on a cheap phone. A deep-learning model trained on Pakistani crop images flags whitefly early. The farmer sprays only the affected block, saves pesticide cost, and protects yield — which matters because agriculture is a large part of Pakistan's economy.

### REMEMBER IT

**Five sectors: E-H-A-B-L.** Education, Healthcare, Agriculture, Business/banking, Local languages. Policy words: innovation, skills, ethics, digital growth.

### COMMON MISTAKE

**Mistake:** Writing only "AI will make Pakistan modern" with no sector or example.

**Fix:** Name the National AI Policy 2025, pick two sectors from the table, and give a local example (rural hospital without a specialist; Urdu/Sindhi speech tools; bank fraud).

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#### CHECK YOURSELF

1. Name the national policy mentioned for AI in Pakistan.
2. Why is AI in healthcare especially important for rural areas?
3. List three skills students should learn to prepare for this future.

Answers: 1. National AI Policy 2025. 2. Those areas often lack specialist doctors; AI can help with disease detection and medical image analysis. 3. Any three of: programming (Python), mathematics, data analysis, problem-solving, ethical use of technology.

Next lecture: 5.2 Applications of Neural Networks and Deep Learning